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Studierichting: Informatica
Oefeningenreeks 3A nr. 17.3

$$f(x) = \frac{4x+5}{6x+7}$$

$$\text{Quotiregel: } \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Stel $u = 4x + 5$ en $v = 6x + 7$

$$u' = \frac{d}{dx}[4x + 5] = 4$$

$$v' = \frac{d}{dx}[6x + 7] = 6$$

$$\begin{aligned}\frac{df}{dx} &= \frac{(6x+7) \cdot \frac{d}{dx}[4x+5] - (4x+5) \cdot \frac{d}{dx}[6x+7]}{(6x+7)^2} \\ &= \frac{(6x+7) \cdot 4 - (4x+5) \cdot 6}{(6x+7)^2} \\ &= \frac{4(6x+7) - 6(4x+5)}{(6x+7)^2} \\ &= \frac{24x+28-24x-30}{(6x+7)^2} \\ &= \frac{28-30}{(6x+7)^2} \\ &= \frac{-2}{(6x+7)^2}\end{aligned}$$

References